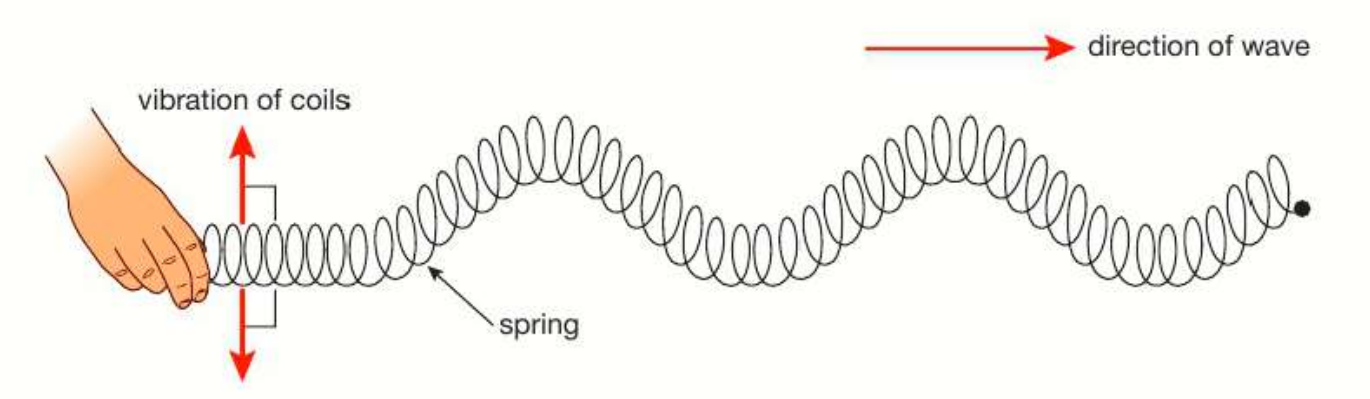
PROPERTIES OF WAVES

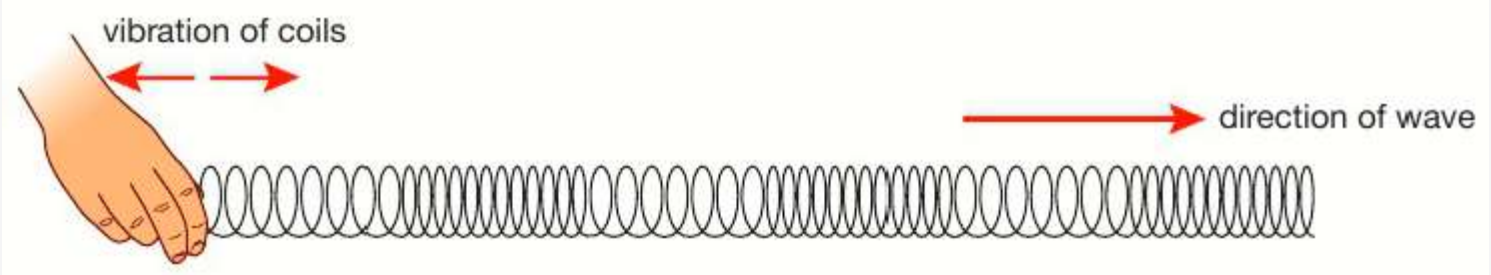
Learning Objectives:

1. *Explain the definition of a wave*
2. *Differ transverse & longitudinal waves*
3. *Define amplitude, wavelength, wavefront, period and frequency of a wave*
4. *Apply the relationship speed, frequency, and wavelength of a wave*
5. *Recall the relationship of frequency & period of a wave*

WAVES

Definition





A wave is a vibration that propagates in a certain direction.

A wave is transferring energy from place to place.

WAVEFRONT

Definition

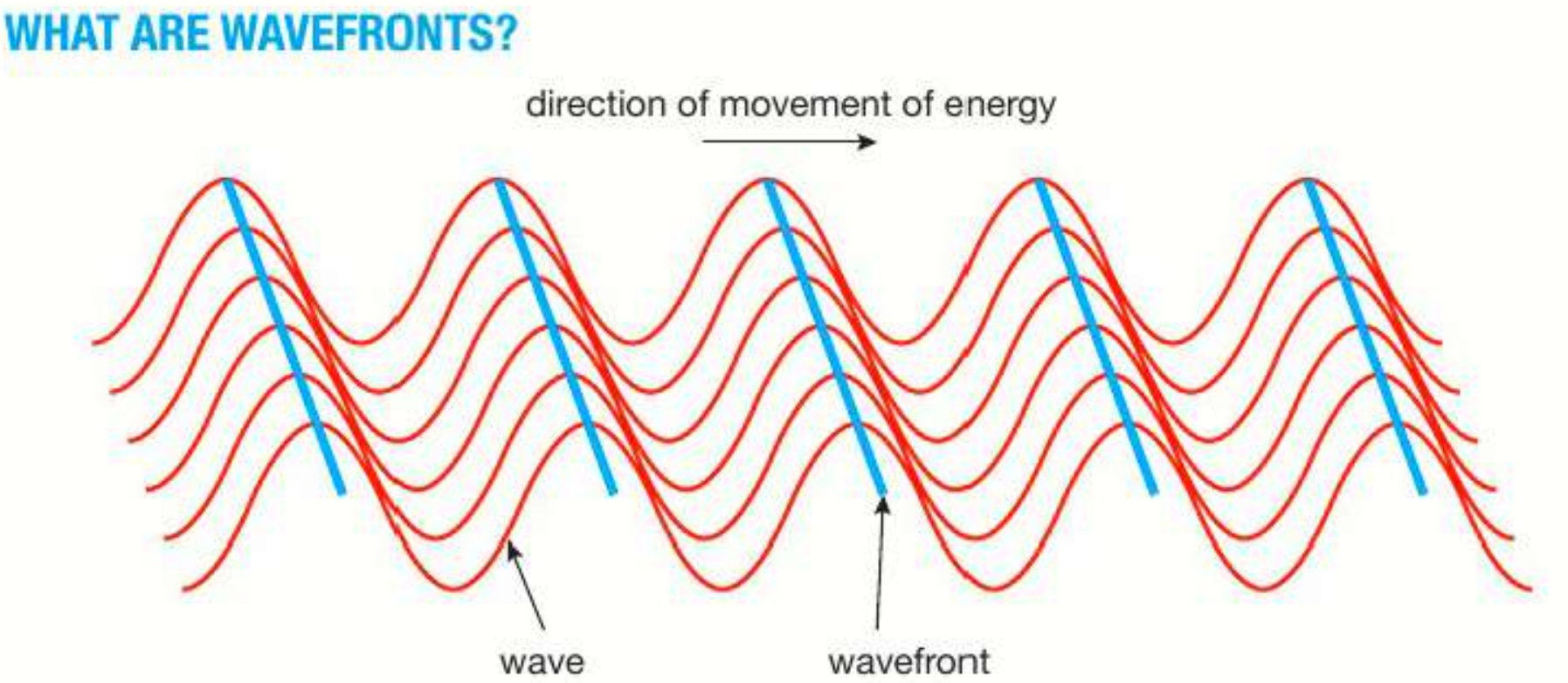
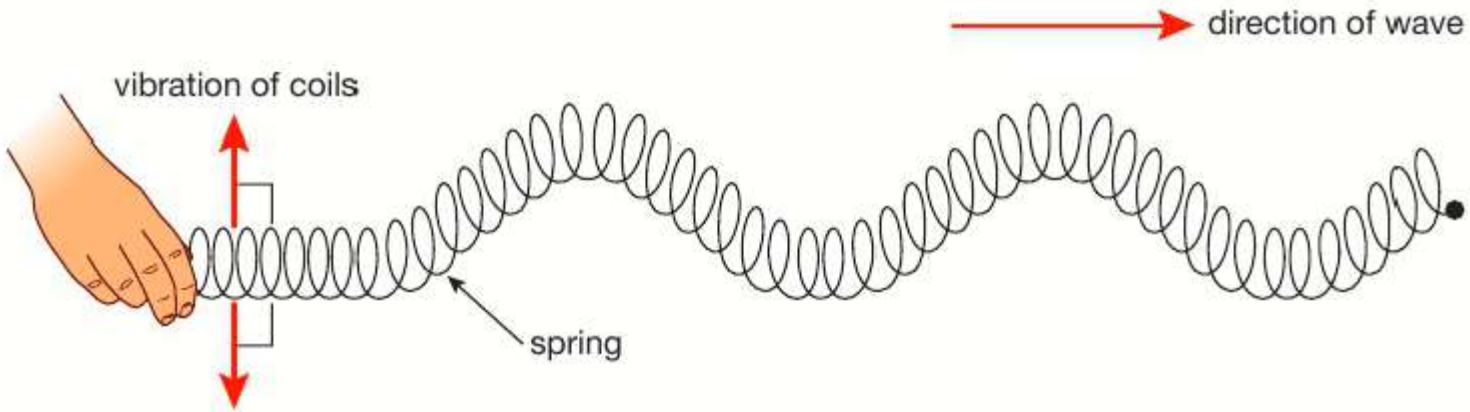


Figure 10.3 Wavefronts are created by overlapping lots of different waves. A wavefront is a line where all the vibrations are in phase and the same distance from the source.

Transverse Waves

Definition

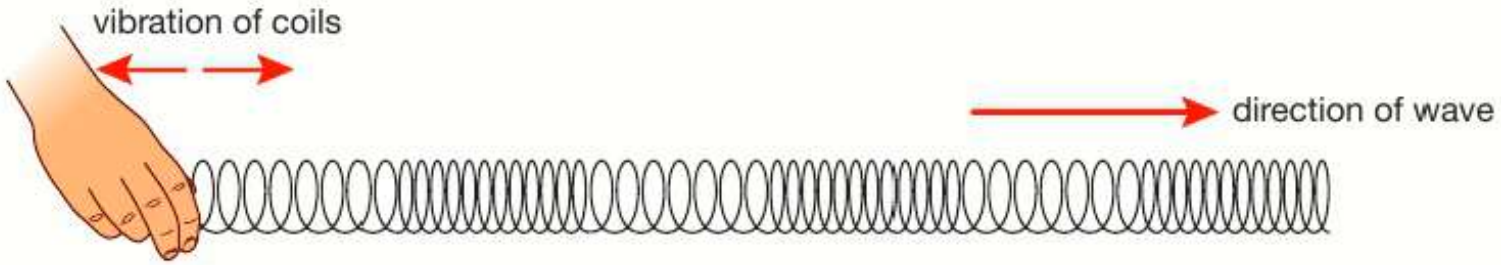
A transverse wave is one that vibrates, or oscillates, at right angles to the direction in which the energy or wave is moving. Examples of transverse waves include light waves and waves travelling on the surface of water.



▲ Figure 10.4 A transverse wave vibrates at right angles to the direction in which the wave is moving.

Longitudinal Waves

Definition



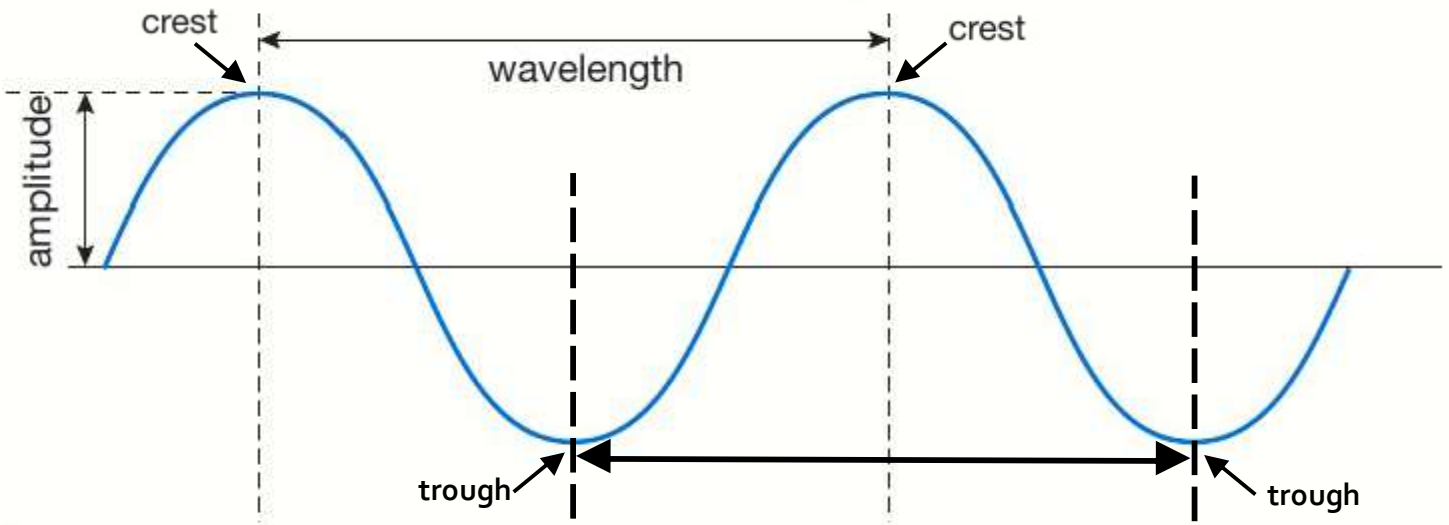
▲ Figure 10.5 A longitudinal wave vibrates along the direction in which the wave is travelling.

A longitudinal wave is one in which the vibrations, or oscillations, are along the direction in which the energy or wave is moving. Examples of longitudinal waves include sound waves.

Amplitude and Wavelength

Recognition

When a wave moves through a substance, its particles will move from their equilibrium (resting position). The maximum movement of particles from their resting or equilibrium position is called its **amplitude** (A).



▲ Figure 10.6 A wave has amplitude and wavelength.

The distance between a particular point on a wave and the same point on the next wave (for example, from crest to crest) is called the **wavelength** (λ).

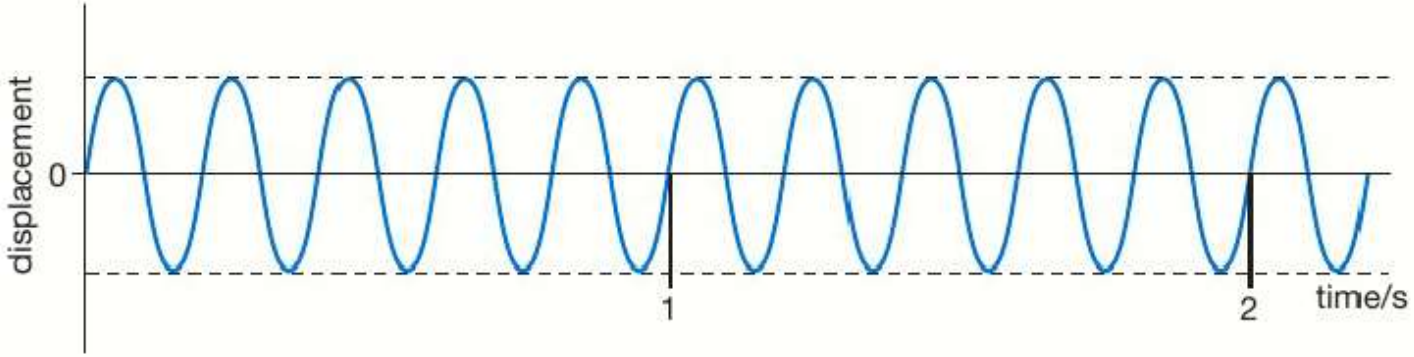
Or the wavelength can be the distance from trough to trough

KEY POINT

λ is the Greek letter lambda and is the usual symbol for wavelength.

Frequency & Period of a wave

Description



▲ Figure 10.8 This graph shows a wave with a frequency of 5 Hz.

The time it takes for a source to produce one wave is called the **time period** of the wave (T). It is related to the frequency (f) of a wave by the equation:

$$\text{frequency, } f\,(\text{Hz}) = \frac{1}{\text{time period, } T\,(\text{s})}$$

$$f = \frac{1}{T}$$

This equation can also be written as

$$T = \frac{1}{f}$$

EXAMPLE 1

Calculate the period of a wave with a frequency of 200 Hz.

$$T = \frac{1}{f}$$

$$= \frac{1}{200\ \text{Hz}}$$

$$= 0.005 \text{ or } 5\ \text{ms } (1000\ \text{ms} = 1\ \text{s})$$

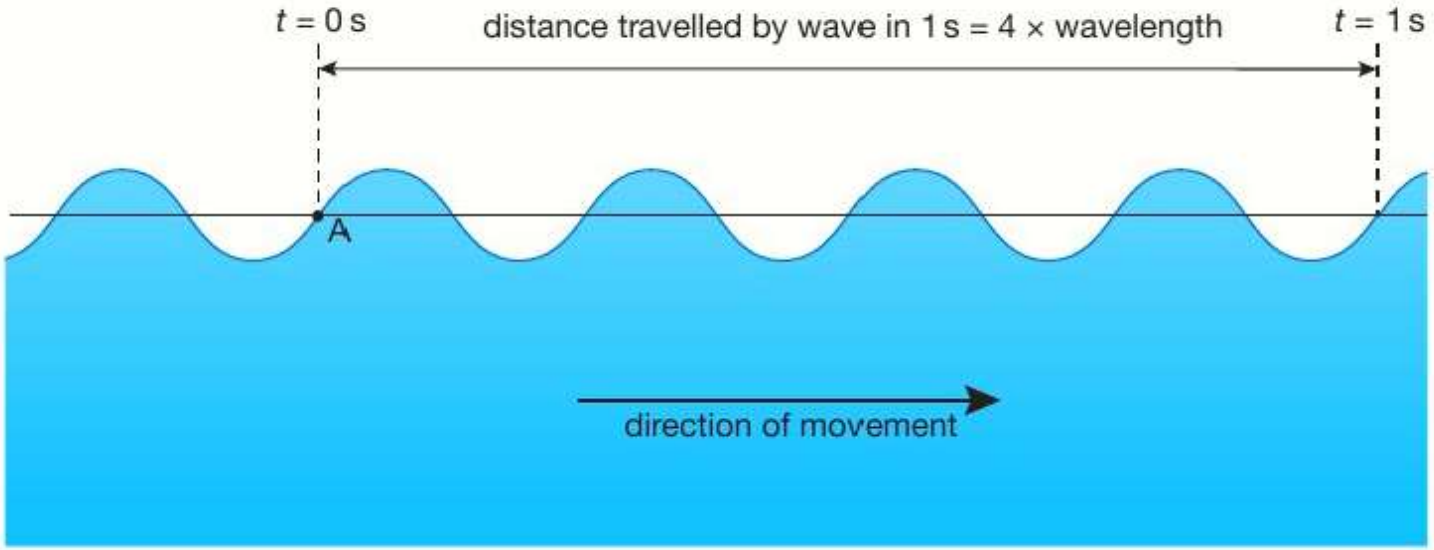
How to Apply the relationship speed, frequency, and wavelength of a wave

THE WAVE EQUATION

There is a relationship between the wavelength (λ), the frequency (f) and the wave speed (v) that is true for all waves:

wave speed, v (m/s) = frequency, f (Hz) × wavelength, λ (m)

$$v = f \times \lambda$$



▲ Figure 10.10 A wave with a frequency of 4 Hz

Imagine that you have created water waves with a frequency of 4 Hz. This means that four waves will pass a particular point each second. If the wavelength of the waves is 3 m, then the waves travel 12 m each second. The speed of the waves is therefore 12 m/s.

$$v = f \times \lambda$$
$$= 4\ \text{Hz} \times 3\ \text{m}$$
$$= 12\ \text{m/s}$$

How to Apply the relationship speed, frequency, and wavelength of a wave

EXAMPLE 2

A **tuning fork** creates sound waves with a frequency of 170 Hz. If the speed of sound in air is 340 m/s, calculate the wavelength of the sound waves.

$$v = f \times \lambda$$

So $$\lambda = \frac{v}{f}$$

$$= \frac{340 \text{ m/s}}{170 \text{ Hz}}$$

$$= 2 \text{ m}$$

▲ Figure 10.11 A tuning fork

Thank You

for your good attention